Another good title would have been *The Immortal Gene.* As well as being more poetic than 'selfish', 'immortal' captures a key part of the book's argument. The high fidelity of DNA copying—mutations are rare—is essential to evolution by natural selection. High fidelity means that genes, in the form of exact informational copies, can survive for millions of years. Successful ones, that is. Unsuccessful ones, by definition, don't. The difference wouldn't be significant if the potential lifespan of a piece of genetic information was short anyway. To look at it another way, every living individual has been built, during its embryonic development, by genes which can trace their ancestry through a very large number of generations, in a very large number of individuals. Living animals have inherited the genes that helped huge numbers of ancestors to survive. That is why living animals have what it takes to survive—and reproduce. The details of what it takes vary from species to species—predator or prey, parasite or host, adapted to water or land, underground or forest canopy—but the general rule remains.

A central point of the book is the one developed by my friend the great Bill Hamilton, whose death I still mourn. Animals are expected to look after not only their own children but other genetic relatives. The simple way to express it, and the one that I favour, is 'Hamilton's Rule': a gene for altruism will spread if the cost to the altruist, *C,* is less than the value, *B,* to the beneficiary, devalued by the coefficient of relatedness, *r,* between them. *r* is a proportion between 0 and 1. It has the value 1 for identical twins; 0.5 for offspring and full siblings; 0.25 for grandchildren, half-siblings, and nieces; 0.125 for first cousins. But when is it zero? What is the meaning of zero on this scale? This is harder to say, but it is important and it was not fully spelled out in the first edition of *The Selfish Gene*. Zero does *not* mean that the two individuals share no genes in common. All humans share more than 99 percent of our genes, more than 90 percent with a mouse, and three-quarters of our genes with a fish. These high percentages have confused many people into misunderstanding kin selection, including some distinguished scientists. But those figures are not what is meant by *r.* Where *r* is 0.5 for my brother (say), it is zero for a *random member of the background population with whom I might be competing*. For purposes of theorizing about the evolution of altruism, *r* between first cousins is 0.125 only when compared to the reference background population ($r = 0$), which is the rest of the population to whom altruism potentially might have been